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Factors motivating and influencing clothing disposal behaviours

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Abstract

This study of 232 college students was framed by Ajzen and Fishbein's theory of reasoned action and examined factors that motivate and influence college students' clothing disposal behaviours. Using survey questionnaires, we investigated four different clothing disposal options (resale, donation, reusing and discarding) and examined five motivational factors (environmental, economic, charity and convenience concerns and information unavailability). The influences of attitudes towards the environment and subjective norms of family and friends on clothing disposal behaviours also were examined. Results indicated that resale and donation behaviours were explained by environmental concerns and that reuse and resale behaviours were explained by economic concerns. Charity concerns motivated donation behaviours and convenience motivated discarding behaviours. Further, findings indicated that family subjective norms influenced environmentally motivated resale and donation behaviours.

Introduction

The US apparel industry generates $345 billion a year, and Americans purchased 20.1 billion garments in 2007 (AAFA, 2007). Despite rising cost pressures for manufacturers and importers, retail prices for apparel have fallen due to the availability of cheap clothes from low-wage countries. Further, globally oriented 'fast fashion' suppliers (e.g. H&M, Zara, Top Shop, etc.) have offered low-priced clothing with a short life cycle for young US consumers. The majority of consumers perceive the lifetime of clothing to be shorter than ever before (Morley *et al*., 2006). According to Birtwistle and Moore (2007), a trend of 'throwaway' fashion attitudes is growing among consumers, especially young fashion leaders. The short fashion life cycle and low price have contributed to the growth of unwanted clothing items. On average, individual Americans throw away more than 68 pounds of clothing each year (Claudio, 2007).

Textiles represent one of the untapped consumer commodities with strong reuse and recycling potential (Domina and Koch, 2002). Textile waste consists of pre- and post-consumer materials (Council for Textile Recycling, 1997). Pre-consumer textile waste (factory waste) is generated during textile manufacturing and is recycled into new raw materials. Post-consumer textile waste consists of used garments and household articles, which can be reused but typically are disposed of into the trash which goes to municipal landfills. In 2009, textiles accounted for about 13 million tons of the solid waste but only 15% are recovered (US EPA, 2009).

Globally, consumers are increasingly concerned about the environment and are willing to participate in sustainable consumption such as purchasing green products and recycling (Claudio, 2007). World Business Council for Sustainable Development (WBCSD) (2008) reported that 96% of Europeans said protecting the environment is important for them personally, and one in four US adults paid attention to environmental issues. Although US consumers are concerned about the environment and participate in waste reduction, the overall bulk of municipal solid wastes continues to rise (US EPA, 2009).

Recycling textiles reduces not only the volume in landfills but also the use of material resources such as fibres, water and chemical dyestuffs (Council for Textile Recycling, 1997). Insufficient recovery of post-consumer textile waste is the biggest obstacle to textile recycling. It is critical to promote appropriate textile disposal practices to meet the goals of recovering post-consumer textile waste in order to reduce the environmental impact of textile waste. Further, though consumer awareness of the need for recycling has been increased, textile disposal behaviour of consumers is minimally understood (Birtwistle and Moore, 2007).

According to Solomon and Rabolt (2009, p. 120), motivation refers to 'the processes that lead people to behave as they do'. Understanding consumer motivations for clothing disposal behaviours can provide a basis for developing effective recycling programmes and policies that preserve natural resources. This study focuses on college students majoring in textiles and apparel. We specifically targeted this group of consumers who are

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knowledgeable about fashion and textiles to examine their motivations and behaviours related to the disposal of unwanted clothing. Many graduates of textiles and apparel programmes go on to work in the retail industry and may become important influencers of the consumer behaviour of others. In addition, college students are more fashion-oriented and purchase more 'fast fashion' clothing than other consumer groups (Seckler, 2005; Birtwistle and Moore, 2007). It has been estimated that college students spend about $6 billion annually on clothing and footwear in the US, which accounts for about 10% of their total purchases (Seckler, 2005; Crane, 2007).

Although college students' clothing consumption is significant, only a few studies have investigated their clothing disposal behaviours. In addition, it is believed that, like purchasing behaviour, consumer disposal behaviour is influenced by reference groups such as family and friends. According to Ward (1974), young children learn consumer behaviour in terms of skills, knowledge and attitudes from their parents and friends. Studies have reported influences of family and peers on decision making of young consumers (e.g. Lachance *et al*., 2003; Garrison and Gutter, 2010). In the near future, young college students may become those parents who model disposal behaviours for their children. Understanding the clothing disposal motivations and behaviours of this group is important for developing strategies to promote environmentally friendly clothing disposal activities.

The purpose of this study was to identify factors that motivate clothing disposal behaviours and to examine the influences of environmental attitude and family and friends on environmentally motivated disposal behaviours. Findings could contribute to the development of educational efforts and community recycling programmes based on consumer motivations to dispose of textiles appropriately.

Literature review and hypotheses

Consumer attitudes towards environmentalism, disposal behaviour and influence of reference groups on consumer behaviours are reviewed to identify factors that motivate and influence clothing disposal behaviours. Hypotheses are developed based on Ajzen and Fishbein's (1977) theory of reasoned action.

Consumer attitude towards environmentalism

Stern (2000, p. 5) defined environmentalism as 'the propensity to take actions with pro-environmental intents'. Pro-environmental behaviour includes both purchase and disposal behaviours in consumption. Environmental purchasing includes buying environmentally friendly or 'green' products, whereas environmental disposal behaviours focus on reduction in household waste and promotion of recycling and reuse (Belli, 2007).

Although researchers have examined the relationship between attitudes towards environmentalism and a broad range of pro-environmental behaviours (Schwartz, 1977; Stern, 2000; Aoki, 2005), few studies have focused specifically on textile disposal (Shim, 1995). In a study of textile recycling behaviours, Koch and Domina (1997) found that general environmental attitudes positively influenced textile recycling. Shim (1995) also examined the influence of consumer attitude towards environmentalism on disposal behaviours and found that environmental attitudes influenced environmentally oriented clothing disposal behaviour. Positive attitude about the environment also significantly influenced pro-environmental apparel purchasing behaviours (Butler and Francis, 1997). Similarly, Hines and Swinker (1996) found that purchase intention of textiles made of recycled materials is influenced by pro-environmental attitudes, but its effect diminishes when a higher price is involved.

Clothing disposal behaviour

Consumers dispose of clothing for a number of reasons such as poor fit, outdated style, boredom and/or wear-out (Koch and Domina, 1999). In order to get rid of their no-longer-needed textiles, consumers have several options: discard, donate, reuse, trade or sell (Solomon and Rabolt, 2009). Shim (1995) interviewed college students and identified eight clothing disposal motivations and linked each to specific clothing disposal behaviours: (1) economically motivated resale; (2) environmentally motivated resale; (3) charity-motivated donation; (4) environmentally motivated donation; (5) economically oriented reuse; (6) environmentally motivated reuse; (7) convenience-oriented discarding; and (8) unawareness-based discarding. While these behaviours were separately examined and linked to their primary motivation, Shim (1995) noted that there may be multiple motivations that drive certain disposal behaviour. One individual may choose resale not only to get economic return but also to preserve the environment. Further, it is typical that a consumer engages in more than one disposal behaviour (Shim, 1995).

Koch and Domina (1999) found that donation to non-profit organizations and passing on to family and friends are common options for clothing disposal. In their study, respondents indicated that the most frequently used charities for clothing items were the Salvation Army and Goodwill. Consistent with purposes of such organizations, altruistic concern (e.g. helping others and raising money for a cause) is the primary motivation of donations among consumers (Shim, 1995; Koch and Domina, 1999).

Selling of used goods for economic benefits is well understood. Studies have found that monetary incentives or rewards promote pro-environmental behaviours such as recycling and reusing. For example, Jacobs and Bailey (1982) found that any type of incentive, especially monetary incentives, increased consumer recycling behaviour. Unwanted garments may be sold directly to other consumers through consignment/vintage stores or online to generate income (Rucker *et al*., 1995). Claudio (2007) reported that about 12%–15% of Americans shop at resale stores. Due to recent popularity of online shopping, consumers use online auctions (e.g. eBay), which allows selling goods (e.g. vintage, used, new and even celebrity clothing items) directly to other consumers. According to Neislsen ratings, eBay's clothing site has more visitors than Victoria Secret's (Lee, 2007). Recently, clothes swapping and swishing are becoming popular and allow another way consumers participate in clothing recycling. Swishing and swap web sites (e.g. swishing.com) allow consumers to exchange valued but no-longer-used clothing with other consumers.

Reuse includes converting textiles for different purposes. Post-consumer textile waste has been collected and recycled through reprocessing of fibres or converting them into different products (e.g. paper) (Claudio, 2007). In 2005, Patagonia launched a Common Threads Recycling Program, which has encouraged the

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apparel industry to participate in recycling. For example, polyester or nylon fibres from used clothing are decomposed in chemical solutions and converted into new fibres (Claudio, 2007). A challenging situation for the textile recycling industry, however, is the abundance of low-priced products, especially from low-wage countries (Birtwistle and Moore, 2007). It has been noted that low-priced low-quality garments do not provide adequate justification for recycling and are difficult to recycle (Britten, 2008). Cheap garments are not of high enough quality to be recycled, and collection costs typically exceed revenue generated (Morley *et al*., 2006). Further, the cost of reverse transportation and recycled fibre production is high, though, so, few companies have participated.

Researchers have noted that participation in recycling programmes is related to convenience and accessibility (Folz, 1991; Domina and Koch, 2002). Derksen and Gartrell (1993) found that communities with access to recycling programmes had a higher level of participation in recycling. A national survey on recycling coordinators reported that communities with a curbside collection programme for recyclable solid waste had a 24% higher participation rate than communities with a drop-off site (Folz, 1991). Convenient access to recycling programmes significantly increased participation in household recycling because consumers perceived that convenience reduced time spent recycling (Halvorsen, 2008). A survey of Goodwill industries also found that people would not go more than 10 min out of their way to make a drop-off (Solid Waste District of LaPorte County, 2008).

Koch and Domina (1999) found that convenience and familiarity with the recycling process were the reasons consumers chose a textile disposal option. They also reported that consumers did not participate in recycling textile waste because current recycling programmes did not include textiles, and they liked to avoid hassles.

Researchers have examined the relationship between disposal behaviours and recycling information availability. Information about a recycling programme significantly increases the likelihood of consumer participation in the programme (De Young, 1989; Domina and Koch, 2002). De Young (1989) compared recyclers and non-recyclers and found that they were not significantly different in their attitudes and views towards recycling but were different in the amount of information needed about how to carry out the recycling activities. In a study of convenience and frequency of recycling, Domina and Koch (2002) found that respondents indicated they did not participate in recycling because they did not have local recycling programmes and did not know other options.

Theoretical framework

It is clear that peers and family constitute a reference group influencing young consumers for both purchasing and disposing behaviours. According to Foxall *et al*. (1998, p. 124), '. . . any individual or collection of people whom the individual uses as a source of attitudes, beliefs, values, or behavior. . . and whose social perspectives are assumed by the individual as a frame of reference for his or her own actions'.

Influence of reference groups on consumer behaviour is well explained by Ajzen's theory of reasoned action (Ajzen and Fishbein, 1977). According to the theory, consumer behaviour is influenced not only by self-beliefs but also subjective norms that consist of normative beliefs of family and friends and motivation to comply. The normative belief refers to one's beliefs that others (e.g. family and friends) believe an action should or should not be taken, and motivation to comply refers to the degree to which one wishes to consider others' beliefs before taking action. Together, beliefs, attitudes and subjective norms are linked to intentions and subsequent behaviours (Ajzen and Fishbein, 1977). In this study, the theory of reasoned action was used as a framework to examine influences of consumers' belief about environmental issues and motivation to comply with family and friends' recycling behaviours on clothing disposal.

Several studies have demonstrated the theory's value in predicting recycling behaviour. For example, Aoki (2005) examined effects of social norms on recycling behaviour and suggested that social norms play a significant role in guiding recycling behaviour. Other studies have emphasized the importance of social norms and the role of context in pro-environmental behaviours. For example, the altruistic behaviour model emphasizes social norms as a basis for personal norms and beliefs, which in turn influence altruistic behaviour (Schwartz, 1977; Stern, 2000). According to the norm activation theory, individual norms are developed from social norms regarding moral and desirable behaviours (Schwartz, 1977). Social and moral norms affect recycling decisions through feelings of self-respect, guilty conscience and 'warm-glow' (giving in order to feel good about helping others) as well as community efforts/sanctions (Halvorsen, 2008). Pro-environmental attitudes are formed by individuals' belief structures and values related to human–environment relations, consequences and responsibility of corrective action, which are influenced by norms (Stern, 2000).

Based on a review of previous research and the theory of reasoned action (Ajzen and Fishbein, 1977), the following hypotheses were developed.

H1: Clothing resale behaviour is influenced by environmental, economic and convenience concerns.

H2: Clothing donation behaviour is influenced by environmental, economic and charity concerns and by information unavailability.

H3: Clothing reuse behaviour is influenced by environmental and economic concerns and by information unavailability.

H4: Clothing discarding behaviour is influenced by information unavailability and convenience concern.

H5: Environmental attitudes and subjective norms of family and friends influence environmentally motivated clothing resale behaviour.

H6: Environmental attitudes and subjective norms of family and friends influence environmentally motivated clothing donation behaviour.

Method

A self-administered survey questionnaire was developed and administered in classroom settings to the population of textiles and apparel students ($n = 232$) at a Southeastern US university. The majority of respondents were ages 19–23 years (88%), female (92%), never married (96%) and Caucasian (88%). See Table 1.

Questionnaire items measured the degree of participation in specific clothing disposal options, motivational factors, environmental attitudes, subjective norms of family and friends and

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Table 1 Characteristics of the respondents

Variables	Frequency	Percentage
Gender ($n = 232$)		
Male	19	8.2
Female	213	91.8
Age ($n = 232$)		
18–20	120	51.7
21–25	108	46.6
26–30	3	1.3
31 or older	1	0.4
Race ($n = 232$)		
Caucasian	203	87.5
African American	15	6.5
Asian/Pacific islander	11	4.7
American Indian	1	0.4
Other	2	0.9
Marital Status ($n = 231$)		
Single, never married	223	96.1
Married	7	3.0
Divorced	1	0.4
Academic level ($n = 232$)		
Freshmen	30	12.9
Sophomore	65	28.0
Junior	54	23.3
Senior	78	33.6
Graduate	4	1.7
Uncertain	1	0.4

demographics. All items on composite measures were measured using 5-point Likert-type response choices (1 = *strongly disagree*, 5 = *strongly agree*).

For clothing disposal options, items measuring *resale* addressed selling to second-hand stores; *donation* addressed giving to charity; *discarding* addressed throwing away; *reuse* addressed using for other purposes. To measure motivational factors for each disposal option, various items were developed from the literature (e.g. Shim, 1995; Koch and Domina, 1999; Domina and Koch, 2002): environmental concerns (to decrease landfills), economic concerns (to save money for me), charity concerns (to raise money for charities), information unavailability (there is no information available for other options) and convenience concerns (I do not have time). Within each disposal option, respondents were asked to indicate their level of agreement with motivational factors (e.g. I often sell my unwanted clothing to save money for me).

For measurement of environmental attitudes, a 5-item scale was adopted with permission from Shim (1995). For example, 'Environmental issues are very important to me'. For subjective norm of family and friends, six items were developed based on Ajzen and Fishbein's conceptualization, four items measured normative beliefs of family's (or friends') environmental attitude/behaviour, and two items measured motivation to comply with family (or friends). Examples of normative beliefs were 'my family likes to recycle' or 'my friends like to recycle'. An example of a motivation-to-comply item was 'I like to do things that my family likes to do'.

Reliability coefficients indicated appropriate internal consistency for self-environmental attitude items ($\alpha = 0.78$), normative beliefs of family ($\alpha = 0.86$), normative beliefs of friends ($\alpha = 0.86$), motivation to comply with family ($\alpha = 0.75$) and motivation to comply with friends ($\alpha = 0.77$). The items measuring each construct were averaged for analysis and mean scores could range from 1–5 for each. Higher scores indicated more positive attitudes, beliefs or motivations. The mean score of the environmental attitude scale was 3.60(SD = 0.67), 3.15 (SD = 0.87) for normative beliefs of family, 3.99(SD = 0.84) for normative beliefs of friends, 3.67(SD = 0.82) for motivation to comply with family and 2.84(SD = 0.77) for motivation to comply with friends.

Results

Factors motivating clothing disposal options

A series of multiple regression analyses were performed to examine the influence of motivational factors on each disposal behaviour (see Table 2). For the resale behaviour, the regression model was significant [$R^2 = 0.24$, $F(3, 214) = 42.64$, $P < 0.001$] and explained 24% of the variance. Environmental concerns ($\beta = 0.24$, $P < 0.01$) and economic concerns ($\beta = 0.29$, $P < 0.01$) were significantly related to resale behaviour, while convenience concerns did not contribute to explaining resale behaviour in the model. For donation behaviour, the model was significant [$R^2 = 0.31$, $F(4, 221) = 24.57$, $P < 0.001$] and explained 31% of the variance. Environmental concerns ($\beta = 0.25$, $P < 0.01$) and charity concerns ($\beta = 0.47$, $P < 0.01$) were significantly related to donation behaviours, while economic concerns and information unavailability were not significantly related. For reuse behaviours, the model was significant [$R^2 = 0.32$, $F(3, 214) = 33.00$, $P < 0.001$], and 32% of the variance was explained by the model; however, only economic concerns ($\beta = 0.53$, $P < 0.01$) were significantly related to reuse behaviours. For discarding behaviours, the model was significant [$R^2 = 0.38$, $F(2, 221) = 68.42$, $P < 0.001$], showing that 38% of the variance was explained. Convenience was significantly related to discarding behaviours ($\beta = 0.58$, $P < 0.01$), but information unavailability was not.

Factors influencing environmentally motivated resale and donation

The effects of self-environmental attitudes and subjective norms of family and friends on the environmentally motivated resale and donation behaviours were examined through a series of stepwise regressions. For subjective norms of family and friends, the score of normative beliefs of family (and friends) was multiplied by the score of motivation to comply with family (and friends) items to represent the subjective norms of family (and friends) respectively. The mean for subjective norm of family was 12.69 ($n = 231$; SD = 4.86), and the mean for subjective norm of friends was 10.58 ($n = 230$; SD = 4.06).

The magnitude of environmentally motivated resale behaviour was calculated from reported resale frequency multiplied by the environmental motivation score for disposal option. The magnitude of environmentally motivated donation behaviour was calculated in the same way. The mean of the environmentally motivated resale behaviour was 5.80 ($n = 220$; SD = 5.40) and 9.51 ($n = 227$; SD = 6.27) for the environmentally motivated donation behaviour.

Hypotheses (H5 and H6) were partially supported. For environmentally motivated resale behaviour, although the regression

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Table 2 Results of regression analyses for motivations predicting clothing disposal behaviours

	Dependent variables											
	Resale ($n = 217$)			Donation ($n = 225$)			Reusing ($n = 217$)			Discarding ($n = 223$)		
Independent variables	B	SE	β	B	SE	β	B	SE	β	B	SE	β
Environmental concern	0.31	0.10	0.24*	0.24	0.07	0.25**	0.08	10	0.07			
Economic concern	0.39	0.11	0.29**	–0.09	0.08	–0.09	0.64	0.11	0.53**			
Charity concern				0.38	0.05	0.48**						
Convenience concern	–0.02	–0.09	–0.02							0.59	0.06	0.58**
Information unavailability				–0.06	0.07	–0.05	–0.05	0.09	–0.04	0.12	0.08	0.09
R^2	0.24			0.31			0.32			0.38		
Adjusted R^2	0.23			0.30			0.31			0.38		
F	21.96			24.57			33.00			68.42		

*$P < 0.01$; **$P < 0.001$.

Table 3 Results of stepwise regression analysis for variables predicting environmentally motivated clothing disposal behaviours

	B	SE	β	R^2	Adjusted R^2	F
Environmentally motivated resale ($n = 214$)						
Model 1				0.06	0.06	13.97
Subjective norm of family	0.28	0.07	0.24***			
Environmentally motivated donation ($n = 221$)						
Model 1				0.10	0.09	23.01
Environmental attitude	2.87	0.60	0.31***			
Model 2				0.12	0.11	14.99
Environmental attitude	2.07	0.67	0.22**			
Subjective norm of family	0.23	0.09	0.18*			

*$P < 0.05$; **$P < 0.01$; ***$P < 0.001$.

model was significant [$R^2 = 0.06$, $F(1, 213) = 13.97$, $P < 0.001$], only the family subjective norm ($\beta = 0.24$, $P < 0.001$) was included (see Table 3). Environmental attitude and subjective norm of friends did not contribute and were removed from the model. For the environmentally motivated donation behaviour, the model was significant [$R^2 = 0.12$, $F(2, 219) = 14.99$, $P < 0.001$]. Environmental attitude ($\beta = 0.22$, $P < 0.01$) and family subjective norm ($\beta = 0.18$, $P < 0.05$) were significantly related to the environmentally motivated donation behaviour, but the model excluded subjective norm of friends.

Discussion

Findings indicated that economic concerns predicted both resale and reuse behaviours; that is, consumers want to save money. This finding is consistent with the findings of Rucker *et al*. (1995), but Morley *et al*. (2006) indicated that consumers were less likely to participate in resale and reuse of clothing with low value and quality. In addition, consumers' motivation to sell clothing is determined by the value to others of their unwanted garments (Koch and Domina, 1999; Claudio, 2007). Young fashion leaders, such as textiles and apparel graduates are moving towards purchasing 'fast fashion' clothing, and the throwaway fashion attitude is growing (Birtwistle and Moore, 2007). Such consumers may become discarders rather than recyclers. Because prices of new garments imported from low-wage countries are cheaper than the cost of recycling, textile recycling has been discouraged. To provide additional options for such consumers, the fashion and the recycling industry could develop team-based recycling policies and programmes emphasizing economic benefits such as buy-back programmes to activate economic motivation. Currently, some retailers (e.g. Patagonia, Elder Beerman) already are engaged in such programmes; a promotional discount is given if customers bring their no-longer-used textiles to the store. Governments should support an industry-wide adoption of buy-back programmes and offer funding for developing advanced recycling techniques and appropriate policies to support recycling.

As expected, donation was related to charity concerns, and discarding behaviour was related to convenience. These findings confirm previous research about reasons for engaging in specific disposal behaviours (Shim, 1995; Domina and Koch, 2002). About 20% of collected, wearable clothing is sold in second hand stores, and the rest is exported to third world countries (Claudio, 2007). The sales by the charitable second hand stores are strong (e.g. $1.8 billion by Goodwill Industries in 2006) and increasing because fashion-conscious consumers frequently purchase updated clothing and donate outdated styles (Claudio, 2007). In order to promote donation and prevent discarding unwanted textiles, the recycling industry should place drop-off sites or community collection bins in convenient places for easy access such as grocery stores and gas stations.

An interesting finding of this study was that information unavailability of other disposal options was not related to any

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disposal behaviour. The finding is inconsistent with previous findings (De Young, 1989; Domina and Koch, 2002). Today's consumers are aware of various clothing disposal options, yet may choose to discard textile products anyway. In order to prevent discarding behaviours, it is necessary to develop a culture of recycling (e.g. behavioural norms) during early childhood stages of the life cycle.

In this study, both environmentally motivated donation and resale behaviours were influenced by subjective norms of family but not friends. As noted earlier, children learn consumer behaviour by observing and imitating their family members (Ward, 1974). Many of today's young college students will become parents who can influence the recycling behaviours of the next generation. Higher education should include the importance of preserving natural resources by emphasizing the benefits of engaging in recycling. For example, textile companies (e.g. Patagonia) have extracted polymers from either soda bottle or plant-based materials (e.g. corn by-products) and manufactured textiles for use in new clothing (Claudio, 2007). Such practices save natural resources.

Findings indicated that attitude towards the environment was related to donation but not to resale behaviours. Also, neither donation nor resale behaviours were influenced by norms of friends. Although peer influence has been a significant predictor of purchase behaviours (Lachance *et al*., 2003), such influence was not related to disposal behaviours. It may be that environmental motivations are learned through the family environment and attitudes about the environment encourage altruistic environmental behaviours regardless of economic benefits.

With increased consumption of textiles, partly due to rapid changes in fashion and low prices for new garments, the need for textile recycling is increasing. Birtwistle and Moore (2007) noted the negative environmental consequence of fast fashion change and short-lived styles that have created a throwaway trend. When consumers perceive their recycling efforts have been effective in preserving the environment, they are more likely to engage in pro-environmental behaviours (Stern, 2000). Textile donation and recycling participants should be informed about the positive impact of their recycling efforts on the environment.

In conclusion, consumers dispose of unwanted clothing in different ways based on different motivational factors. That is, those who want to save money resell or reuse garments they no longer need, and their families influence their resale behaviours. Those motivated by convenience tend to discard unwanted textiles. Consumers who donate used clothing are motivated by environmental and charity concerns and are influenced by their families in this regard. Educational programmes that promote positive environmental behaviours can be tailored to specific groups and desired textiles disposal options based on these findings.

This study has limitations and suggestions for future studies. The sample was limited to textiles and apparel students from only one university which limits generalizability of the findings. While part of our intent was to determine the motivations and behaviours of this fashion-knowledgeable population, it is possible that because of sensitivity to clothing style changes, their disposal activities may be different from that of other consumer groups. Additional studies should include diverse samples. Cross-sectional surveys of community members would provide findings generalizable to consumers in other age and interest groups. In addition, this study was limited to respondents' attitudes about the environment, and the influence of family and friends on clothing disposal behaviours. Like purchase behaviours, textile disposal behaviours are influenced by other sources, such as mass media efforts and direct marketing. Recently, for example, celebrity endorsers have become powerful reference groups influencing consumer behaviours in general. Future studies should investigate additional influence on clothing disposal behaviours.

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